SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

Project Report on

"College FAQ ChatBot"

submitted in partial fulfillment of the requirement for the completion of First semester of **Innovation and Design Thinking Laboratory (SDCCS1)**

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE and ENGINEERING

(AI&ML)

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DEPARTMENT OF COMPUTER SCIENCE and ENGINEERING

2025-26

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CERTIFICATE

Certified that the project work entitled **"COLLEGE FAQ CHATBOT"** is a bonafide work carried out by JYOTHI PATEL, NIKHIL VERMA, PRATHIKSHA L, SATYA PRAKASH, SUPREET WALE in partial fulfillment for the completion of First semester of Innovation and Design Thinking Laboratory of **BACHELOR OF ENGINEERING** in **COMPUTER SCIENCE and ENGINEERING (AI &ML)** from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26.

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Course Outcomes

CO1: Identify different modes of thinking to understand the problem instead of finding answers/solutions for questions/problems

CO2: Acquire abductive reasoning to find new problems

CO3: Apply the concepts of design thinking concepts to develop a solution to the problem.

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

POs: PO1: Engineering Knowledge, PO2: Problem analysis, PO3: Design/Development of solutions, PO4: Conduct investigations of complex problems, PO5: Modern tool usage, PO6: Engineer and the world, PO7: Ethics, PO8: Individual and collaborative team work, PO9: Communication, PO10: Project management and finance, PO11: Lifelong learning

PROGRAM SPECIFIC OUTCOMES (PSOs):

1. PSO1: Computer based systems development: Ability to apply the basic knowledge of database systems, computing, operating system, digital circuits, microcontroller, computer organization and architecture in the design of computer based systems.

2. PSO2: Software development: Ability to specify, design and develop projects, application softwares and system softwares by using the knowledge of data structures, analysis and design of algorithm, programming languages, software engineering practices and open source tools.

3. PSO3: Computer communications and Internet applications: Ability to design and develop network protocols and internet applications by incorporating the knowledge of computer networks, communication protocol engineering, cryptography and network security, distributed and cloud computing, data mining, big data analytics, ad hoc networks, storage

Abstract

This project is designed to address the growing need for instant, accurate, and user-friendly access to institutional information at Siddaganga Institute of Technology, Tumakuru. Students and visitors often face difficulty in obtaining quick responses regarding admissions, courses, campus life, placements, and facilities through traditional inquiry methods. To overcome this limitation, a web-based FAQ Chatbot with an integrated feedback system has been developed to provide real-time assistance and improve overall user experience.

The main objective of this project is to develop an interactive chatbot that can answer frequently asked questions related to SIT Tumakuru efficiently and reliably. The system aims to reduce manual workload, enhance accessibility of information, and provide categorized query handling for better user navigation. Additionally, a feedback and admin panel is included to collect user opinions and help administrators analyze service quality.

The implementation of the project is carried out using **HTML, CSS, and JavaScript** for front-end development. The chatbot interface includes features such as quick topic selection, search-based question filtering, dark mode, and dynamic chat responses. The feedback system allows users to submit ratings and comments, which are stored in memory and displayed in an admin panel secured with login credentials. The entire system runs as a lightweight web application requiring only a modern web browser for operation

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Chapter 1

Introduction

1.1 Motivation

In today's digital era, students and visitors expect quick and accurate access to information through online platforms. At Siddaganga Institute of Technology (SIT), Tumakuru, a large number of students, parents, and applicants frequently seek information related to admissions, courses, placements, campus facilities, fees, and contact details. Traditionally, this information is obtained through websites, phone calls, or office visits, which is often time-consuming, inconvenient, and dependent on working hours.

Due to the increasing number of queries, it becomes difficult for administration staff to handle repeated questions efficiently. Many users also face difficulty navigating through long web pages to find the exact information they need. This creates the need for an automated, fast, and user-friendly system that can respond instantly to user queries without human intervention.

The motivation behind this project is to develop an intelligent **FAQ Chatbot with a Feedback System** that can provide instant responses to commonly asked questions about SIT Tumakuru. The feedback module further helps in understanding user satisfaction and improving the quality of service. This project aims to reduce manual workload, improve accessibility of information, enhance user experience, and promote digital interaction within the institution.

1.2 Objective of the project

1. To develop a web-based FAQ chatbot for SIT Tumakuru
2. To provide instant and accurate answers to common questions
3. To reduce manual workload of staff
4. To collect user feedback for continuous improvement
5. To improve user experience using simple UI features

Chapter 2

Literature Survey

This chapter presents a review of existing research, tools, and technologies related to **chatbots, web-based information systems, and feedback management systems**. The literature survey provides a foundation for understanding how similar systems have been developed, the technologies used, and the challenges involved. It also highlights how artificial intelligence tools and web technologies are explored and applied in developing automated FAQ systems for educational institutions.

AI Tool Exploration

Artificial Intelligence (AI) has become an essential part of modern software applications, especially in the field of chatbots, automated customer support systems, and virtual assistants. AI enables systems to understand user queries, process information intelligently, and generate meaningful responses. During the initial phase of the **SIT Tumakuru FAQ Chatbot with Feedback System** project, an exploration of various AI-based chatbot development tools and technologies was carried out to identify the most suitable approach for this application.

Sl. No.	Tool / Approach	Type	Key Features	Advantages	Limitations for This Project	Suitability for SIT FAQ Chatbot
1	Google Dialogflow	Cloud-based AI Chatbot	NLP, intent detection, entities, context handling, integration with web/apps	Powerful NLP, easy to build flows, supports multiple languages, rich integrations	Requires internet, API setup, Google Cloud account, may need paid plan for high usage	Medium
2	IBM Watson Assistant	Cloud-based AI Chatbot	AI assistant builder, dialog management, analytics, multi-channel support	Strong enterprise features, analytics dashboard, good for complex business bots	Complex configuration, requires IBM Cloud account, paid beyond free tier	Low–Medium
3	Microsoft Bot Framework	Bot development platform	SDKs, connectors, integration with Azure services and Teams	Highly customizable, integrates well with Microsoft ecosystem, scalable	More coding effort, Azure setup needed, overkill for simple FAQ bot	Low
4	OpenAI / ChatGPT API	Generative AI API	Natural language generation, flexible conversation, handles open-ended input	Very intelligent replies, can handle varied questions, minimal rule-writing	Needs internet & API key, usage cost, responses may vary, not ideal for fixed institutional data	Medium

Table 2.1 AI Tool Exploration

Chapter 3

System Overview

The **SIT Tumakuru FAQ Chatbot with Feedback System** is a web-based application designed to provide instant, reliable, and user-friendly responses to frequently asked questions related to Siddaganga Institute of Technology, Tumakuru. The system is developed to assist students, parents, and visitors by offering quick access to institutional information such as admissions, courses, campus life, placements, facilities, research, fees, and contact details. It also includes a feedback mechanism and an admin panel to support continuous improvement of the system.

The system operates through a simple and interactive **chat-based interface** where users can either select predefined topics or directly type their questions. The chatbot analyzes the user input using a **keyword-based matching algorithm** developed in JavaScript and retrieves the most relevant answer from the predefined FAQ database. This approach ensures high accuracy and fast response time without the need for heavy AI models or cloud-based APIs.

3.1. Stakeholders and Requirements

Category	Role	Primary Interaction with Chatbot
Students	Primary users	Ask academic, administrative, and campus life-related questions
Faculty Members	Secondary users	Provide subject-related FAQs, guide students, monitor queries
Administrative Staff	Supervisors & data users	Manage institutional FAQs, analyze trends, ensure accuracy
IT Support Team	Technical enablers	Implement, maintain, and troubleshoot the chatbot
College Management / Administration (Decision Makers)	Strategic stakeholders	Oversee performance, ensure compliance, and leverage insights for decision-making

Table 3.1 Stakeholders and Requirements

Chapter 4

Functional Requirements

4.1 IPO (Input–Process–Output) of Functions

The system takes user questions and feedback as input. It processes the queries by analyzing and matching them with a predefined FAQ database. The chatbot then provides relevant answers as output and stores user feedback to help improve response quality and system performance over time.

1. User Query Handling (Chatbot Question Answering)

Input:

User enters a question through the chat input box.

Process:

The system searches the FAQ database using keyword matching and relevance scoring. The most relevant answers are identified.

Output:

Correct answers are displayed instantly in the chatbot window.

2. Quick Topic Selection

Input:

User clicks on a topic button (Admissions, Courses, Campus Life, Placements, etc.).

Process:

The system fetches all FAQs related to the selected category from the FAQ database.

Output:

A list of frequently asked questions for that topic is displayed.

3. Browse All Questions

Input:

User clicks on “Browse All Questions”.

Process:

The chatbot displays all FAQs categorized by department with search and filter options.

Output:

Complete categorized FAQ list is shown for easy browsing.

4. Search and Filter Questions

Input:

User enters keywords in the search bar or selects category filters.

Process:

The system filters questions by matching keywords and selected category.

Output:

Relevant questions are displayed while non-matching questions are hidden.

5. Answer Display Function

Input:

User clicks on a specific question.

Process:

The chatbot retrieves the corresponding answer from the FAQ database.

Output:

The selected answer is displayed in the chat window.

6. Typing Indicator Simulation

Input:

User submits a query.

Process:

A typing animation is displayed for a short delay to simulate human-like response.

Output:

Improved user experience with realistic chatbot interaction.

7. Feedback Submission

Input:

User enters name, email, rating, and comments in the feedback form.

Process:

The system validates the input and stores the feedback in memory.

Output:

Success message is displayed confirming feedback submission.

8. Feedback Validation

Input:

Feedback form data.

Process:

System checks whether all required fields are filled correctly.

Output:

Error message shown if validation fails, else feedback is accepted.

9. Admin Login Authentication

Input:

Admin email ID and password.

Process:

System verifies credentials against predefined admin details.

Output:

Admin login success or failure message displayed.

10. Admin Feedback Viewing

Input:

Successful admin login.

Process:

System retrieves stored feedback records.

Output:

Feedback details are displayed in a tabular format in the admin panel.

11. Dark Mode / Light Mode Toggle

Input:

User clicks on Dark Mode or Light Mode button.

Process:

System changes theme variables and saves preference in local storage.

Output:

Interface switches between dark and light themes.

12. 24/7 Accessibility

Input:

User opens the chatbot webpage at any time.

Process:

Web application loads and initializes chatbot services.

Output:

Chatbot is available anytime without restrictions.

4.2 AI Models Used in the System

The SIT FAQ Chatbot uses basic Artificial Intelligence techniques to understand user questions, find relevant answers, analyze feedback, and improve performance. The AI models mainly work on **text data** collected from users and predefined FAQs.

1. Natural Language Processing (NLP) Model – Query Understanding

Type of Input Data:

Text data (user questions typed in the chatbot)

Source of Data:

User queries entered in the chatbot

Predefined FAQ questions and answers stored in the system

Training Algorithm / Tool:

Keyword matching and intent detection

NLP techniques such as tokenization and text comparison

Tools/Concepts: JavaScript-based text processing, NLP logic (can be extended using Dialogflow, Rasa, or GPT)

Description:

This model helps the chatbot understand what the user is asking. It compares the user's question with stored FAQs and finds the most relevant answer.

2. Information Retrieval / Relevance Scoring Model

Type of Input Data:

Text queries and FAQ database content

Source of Data:

FAQ database (Admissions, Courses, Placements, etc.)

User search keywords

Training Algorithm / Tool:

Rule-based relevance scoring

Keyword frequency matching

Sorting based on match score

Description:

This model ranks questions based on how closely they match the user's query and returns the top relevant answers.

3. Personalized Interaction Model

Type of Input Data:

User interaction data (selected topics, repeated queries)

Source of Data:

Chat history during the session

User-selected categories

Training Algorithm / Tool:

Session-based logic

Simple decision rules

Description:

This model adapts responses based on the user's interaction, such as suggesting related questions from the same topic.

4. Feedback Analysis Model (Sentiment-Based)

Type of Input Data:

User feedback text and rating

Source of Data:

Feedback form submitted by users

Ratings (Poor to Excellent)

Training Algorithm / Tool:

Sentiment categorization

Rule-based sentiment identification

Can be extended using ML libraries (e.g., TextBlob, Hugging Face)

Description:

This model analyzes user feedback to understand satisfaction levels and helps improve chatbot responses.

5. Continuous Learning (Conceptual AI Model)

Type of Input Data:

Chat logs and feedback data

Source of Data:

Stored user queries

Feedback and ratings

Training Algorithm / Tool:

Incremental learning (conceptual)

Manual dataset updates by admin

Can be enhanced using ML retraining tools

Description:

The chatbot improves over time by updating FAQs and responses based on repeated questions and user feedback.

6. Summary Table of AI Models

AI Model	Input Data	Data Source	Algorithm / Tool	Purpose
NLP Query Model	User Text Query	Chat Input, FAQ DB	NLP, Keyword Matching	Understand User Questions
Relevance Scoring	Query + FAQ Text	FAQ Database	Rule-Based Scoring	Find Best Answers
Personalization	User Interaction	Chat Session	Decision Rules	Improve User Experience
Feedback Analysis	Feedback Text	Feedback Form	Sentiment Rules	Measure Satisfaction
Continuous Learning	Chat Logs	Admin Updates	Incremental Updates	Improve Accuracy

Table 4.2 Stakeholders and Requirements

Chapter 5

Results

Snapshots

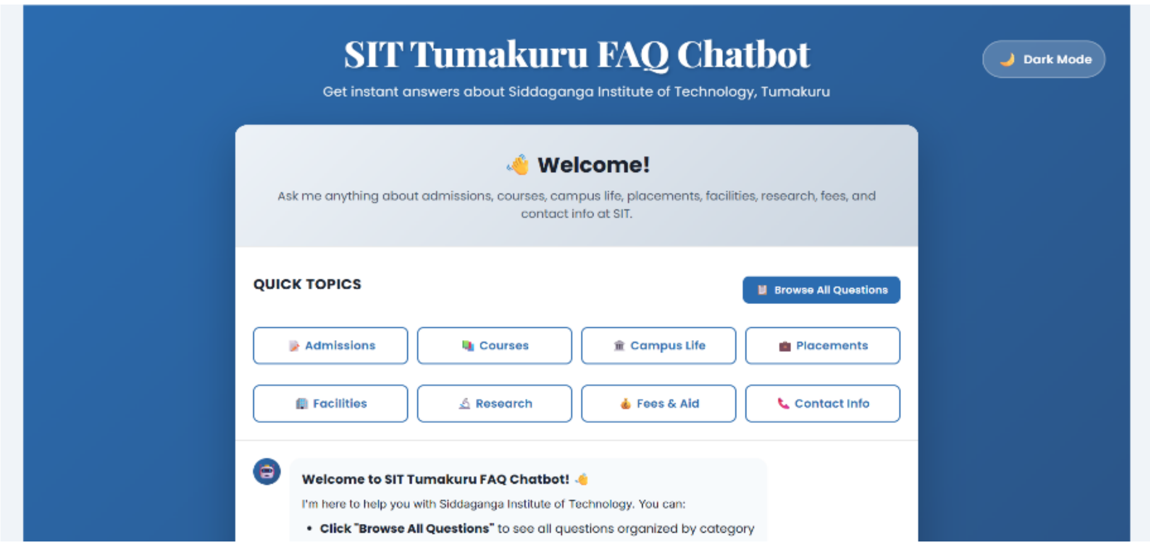

Fig 5.1: Chatbot Home Page

This snapshot shows the main home page of the chatbot, including title, welcome message, and quick topic buttons such as Admissions, Courses, Campus Life, Placements, and Facilities.

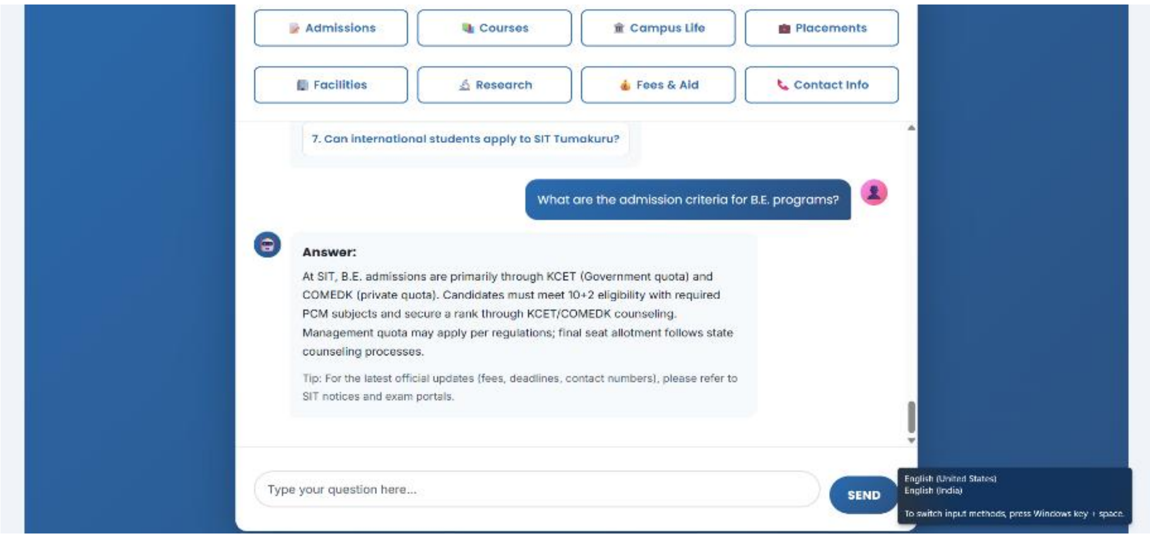

Fig 5.2: User–Chatbot Interaction Screen

This snapshot displays the interaction between the user and the chatbot, where the user enters a query and the chatbot responds with relevant answers.

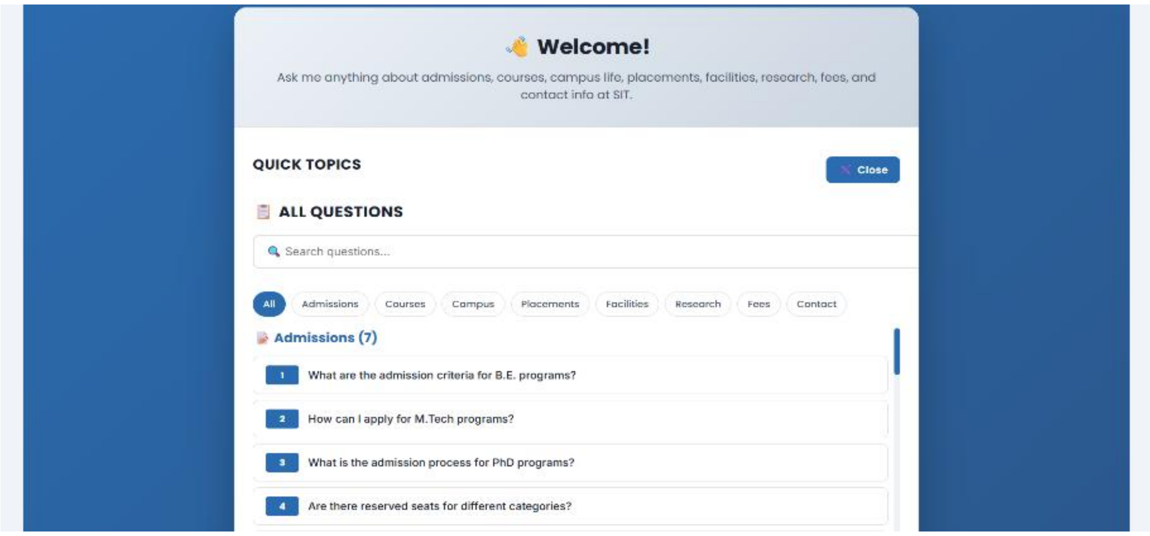

Fig 5.3: Browse All Questions Page

This snapshot shows the “Browse All Questions” feature, where FAQs are displayed category-wise along with search and filter options.

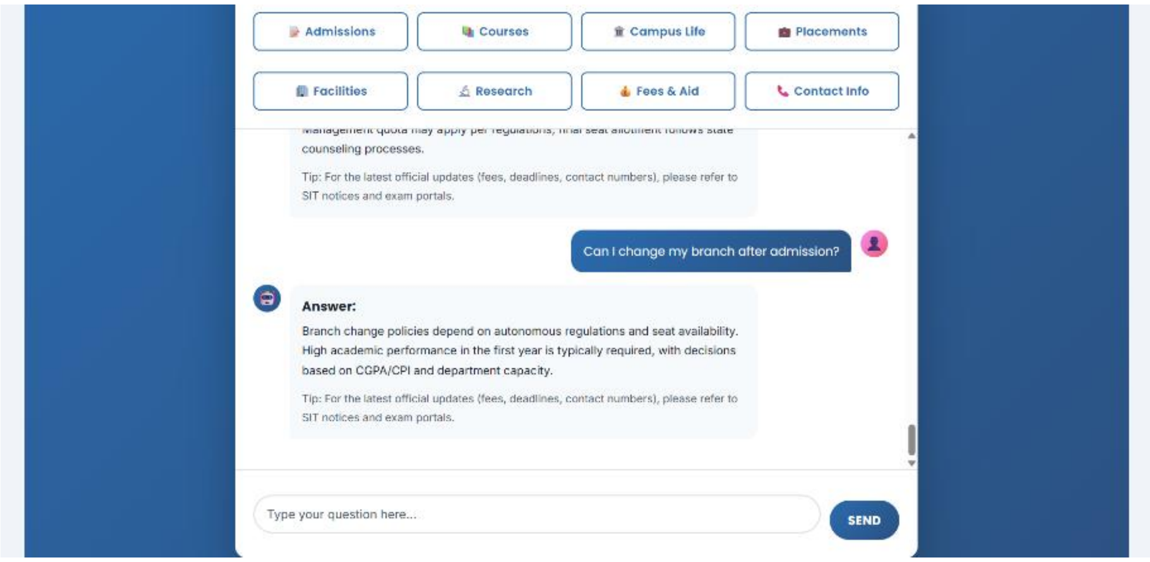

Fig 5.4: Answer Display Screen

This snapshot shows how the chatbot displays detailed answers when a user selects a specific question.

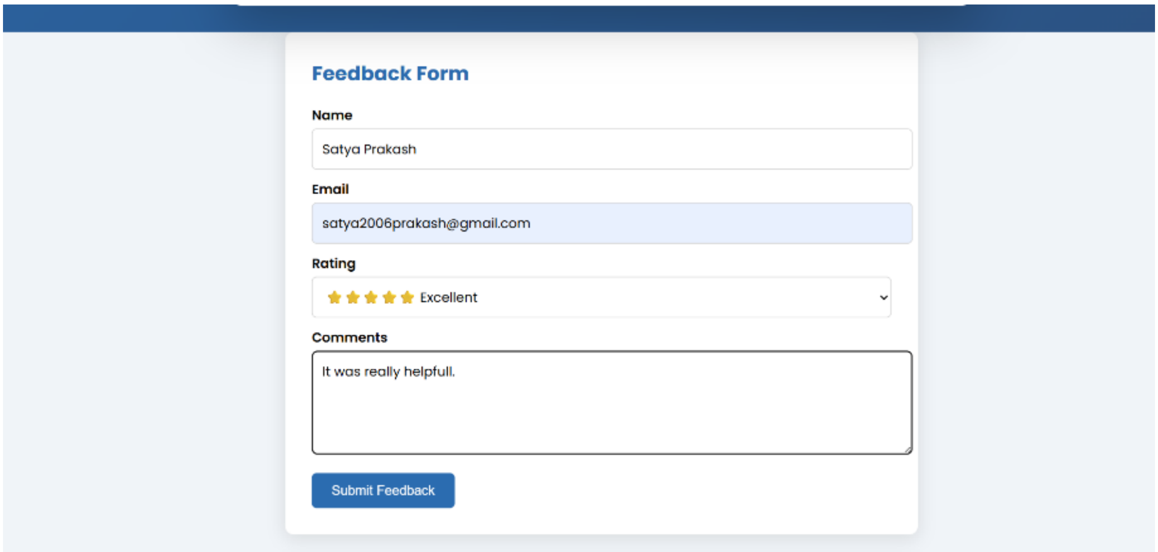

Fig 5.5: Feedback Form Page

This snapshot shows the feedback form where users can enter their name, email, rating, and comments about the chatbot.

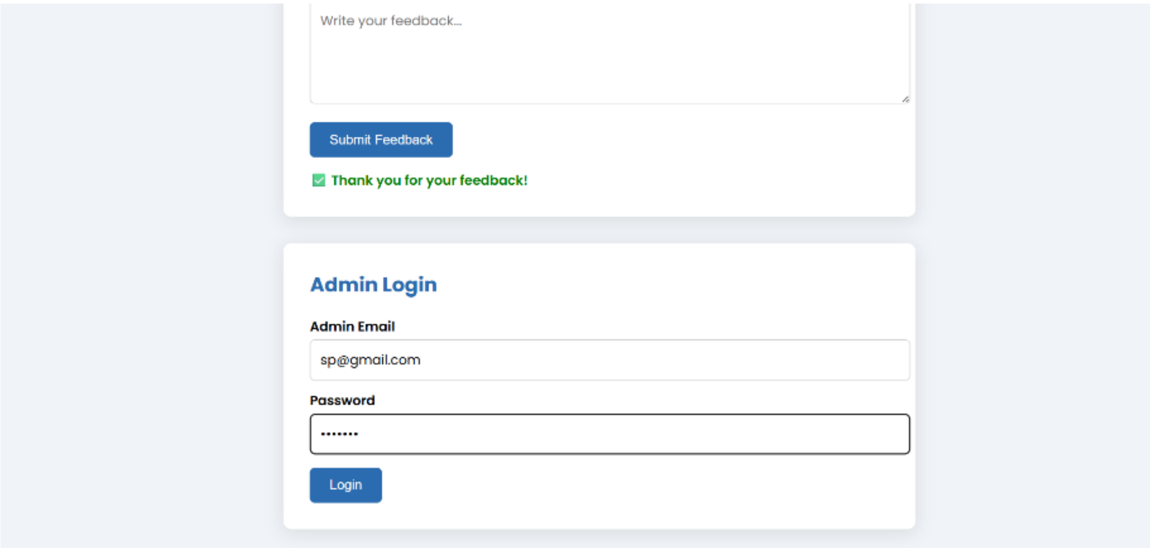

Fig 5.6: Admin Login Page

This snapshot displays the admin login interface, which allows authorized users to access the admin panel.

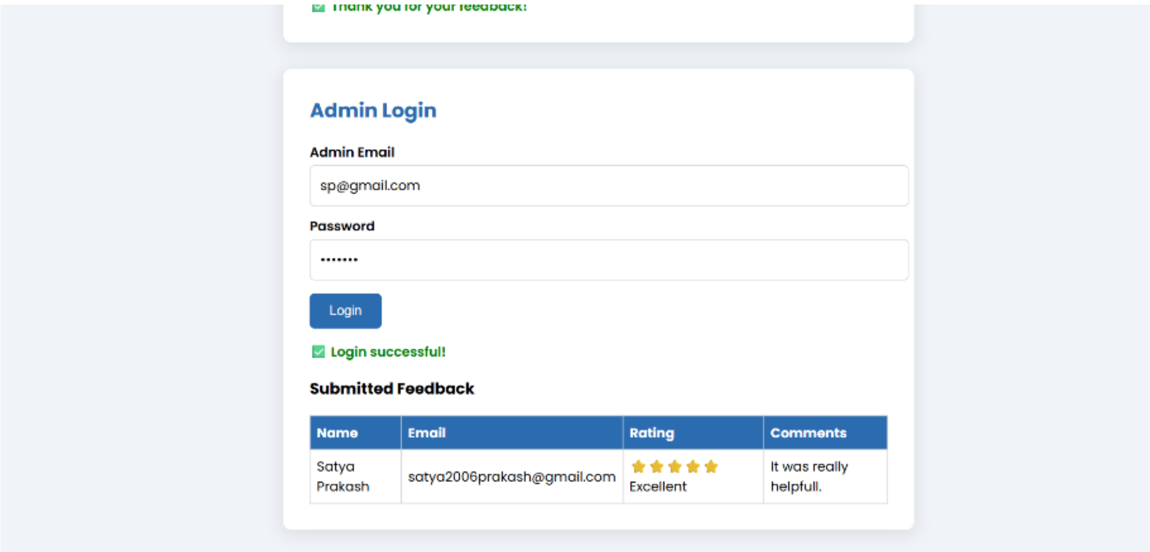

Fig 5.7: Admin Feedback Viewing Panel

This snapshot shows the admin panel where submitted user feedback is displayed in a tabular format.

Fig 5.8: Dark Mode Interface

This snapshot illustrates the dark mode feature of the chatbot, which improves user experience during low-light usage.

Chapter 6

Conclusion

The project titled **“SIT Tumakuru FAQ Chatbot with Feedback System”** has been successfully designed and implemented to provide instant and reliable information related to admissions, courses, campus life, placements, facilities, research, fees, and contact details. The system effectively reduces manual effort by automating the process of answering frequently asked questions through a user-friendly chatbot interface.

The chatbot demonstrates efficient handling of user queries using structured FAQ data, relevance-based matching, and categorized browsing options. Additional features such as quick topic selection, search and filter options, feedback collection, admin panel access, and dark mode enhance usability and user experience. The inclusion of a feedback mechanism allows continuous improvement of the system based on user responses.

Overall, the project fulfills its objectives by offering a 24/7 accessible support system for students and visitors of SIT Tumakuru. The developed system is reliable, easy to use, and scalable, and it can be further enhanced by integrating advanced AI models and real-time database connectivity in the future.

6.1 Scope for future work

The SIT Tumakuru FAQ Chatbot with Feedback System works well for answering common questions, but there is still scope to improve and extend the system in the future. Some features were not included in the current project due to limited time and resources.

In the future, the chatbot can be made more intelligent by using advanced Artificial Intelligence models. This will help the chatbot understand long and complex questions more accurately and give better responses.

The system can also be connected to a live college database so that information such as exam timetables, results, fees, and announcements are updated automatically. This will reduce manual work and keep the information always correct.

Another improvement can be the addition of voice input and voice output, so users can talk to the chatbot instead of typing. This will make the system more user-friendly and helpful for differently-abled users.

The feedback system can be improved by automatically analyzing user feedback to understand whether users are satisfied or not. This will help the administrators improve the chatbot regularly.

In the future, the chatbot can also be developed as a mobile application and integrated with platforms like WhatsApp, so more users can access it easily. These improvements will make the system more powerful and useful in the coming years.

Appendix A

Sustainable Development Goals addressed

#	SDG	Level
1	No Poverty	1
2	Zero Hunger	1
3	Good Health and Well-being	2
4	Quality Education	3
5	Gender Equality	2
6	Clean water and Sanitation	1
7	Affordable and Clean Energy	1
8	Decent work and Economic Growth	2
9	Industry, Innovation and Infrastructure	3
10	Reduced Inequalities	2
11	Sustainable cities and Communities	2
12	Responsible Consumption and production	1
13	Climate action	1
14	Life below water	1
15	Life on Land	1
16	Peace, Justice and Strong Institutions	2
17	Partnership's for the Goals	3

Levels: Poor:1, Good :2, Excellent:3

Appendix B

Self-Assessment of the Project

#	PO and PSO	Contribution from the Project	Level
1	Engineering Knowledge:	Applied basic knowledge of web technologies, programming logic, and computer systems to develop the chatbot.	2
2	Problem Analysis:	Identified the problem of handling repeated student queries and analyzed requirements for an automated solution.	2
3	Design/development of solutions	Designed and developed a working FAQ chatbot with feedback and admin panel features.	3
4	Conduct investigations of complex problems:	Studied user query patterns and tested different scenarios to improve response accuracy.	2
5	Modern tool usage:	Used modern tools such as HTML, CSS, JavaScript, browser storage, and UI design techniques.	3
6	The Engineer and the world:	The project helps students and visitors by providing easy access to college information.	2
7	Ethics:	Ensured ethical use of data by limiting feedback access to admin and avoiding misuse of user information.	2
8	Individual and Team Work:	The project was completed by planning tasks and coordinating design, coding, and testing activities.	2
9	Communication:	Improved communication between students and institution through an automated chatbot system.	3
10	Project Management and Finance:	Managed time, tasks, and resources efficiently without additional financial cost.	2
11	Life-long Learning:	Learned new concepts such as chatbot design, UI development, and basic AI logic beyond classroom syllabus.	3

1	PSO1:	Applied knowledge of computer-based systems and software development to build a functional chatbot.	3
2	PSO2:	Designed and implemented application software using programming concepts and open-source tools.	3
3	PSO3:	Gained understanding of internet-based applications and client-side communication using web technologies.	2

1. PSO1: Computer based systems development: Ability to apply the basic knowledge of database systems, computing, operating system, digital circuits, microcontroller, computer organization and architecture in the design of computer based systems.
2. PSO2: Software development: Ability to specify, design and develop projects, application softwares and system softwares by using the knowledge of data structures, analysis and design of algorithm, programming languages, software engineering practices and open source tools.
3. PSO3: Computer communications and Internet applications: Ability to design and develop network protocols and internet applications by incorporating the knowledge of computer networks, communication protocol engineering, cryptography and network security, distributed and cloud computing, data mining, big data analytics, ad hoc networks, storage

Levels: Poor:1, Good :2, Excellent:3